

Field CAD Onshape link

Score Calculator link

2024 WCP

CADathon Challenge

Game Update 01

Summary

In *WCP DunkTank*, two alliances of three teams each compete to determine who is the dunking GOAT. Teams have 2 minutes and 15 seconds to dunk BASKETBALLS and use ELECTROLYTES to advance their performance. At the end of the match, the alliance with the most points wins!

The first 15 seconds of the match are the AUTONOMOUS PERIOD, where ROBOTS will follow pre-programmed instructions to earn points by completing the following tasks:

1. Dunking BASKETBALLS in HOOPS
2. Scoring ELECTROLYTES in DUNK TANKS
3. Exiting an ALLIANCE ZONE

For the remainder of the match, which is referred to as TELEOPERATED PERIOD, drivers take control of their ROBOTS and earn points by accomplishing the following:

1. Dunking BASKETBALLS in HOOPS
2. Scoring ELECTROLYTES in ALLIANCE DUNK TANKS during the ENDGAME
3. Completely FILLING HOOP STACKS by the end of the match

The final 20 seconds of the TELEOPERATED PERIOD are called the ENDGAME. As explained later on in the manual, specific gameplay elements will change during this period.

The alliance with the highest score at the end of the match wins!

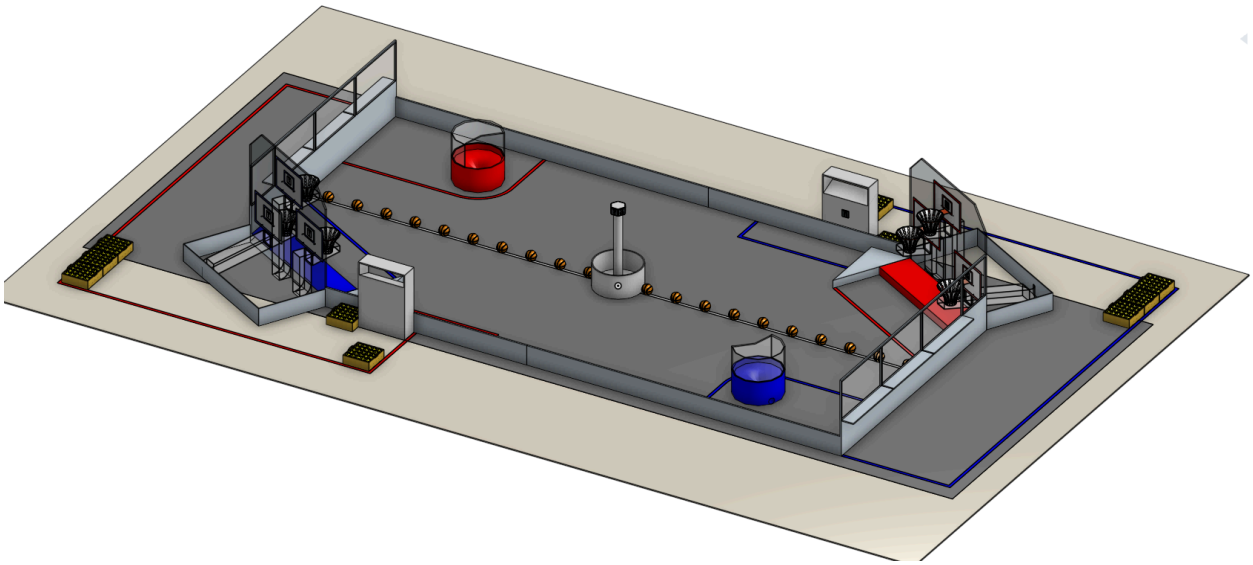
AUTONOMOUS score breakdown

Task	Point Value
Dunk a BASKETBALL in the LOW HOOP	3 points each
Dunk a BASKETBALL in the MIDDLE HOOP	8 points each
Dunk a BASKETBALL in the HIGH HOOP	12 points each
Score an ELECTROLYTE in their ALLIANCE DUNK TANK	½ point each
Exit an ALLIANCE ZONE	6 points per robot

TELEOPERATED score breakdown

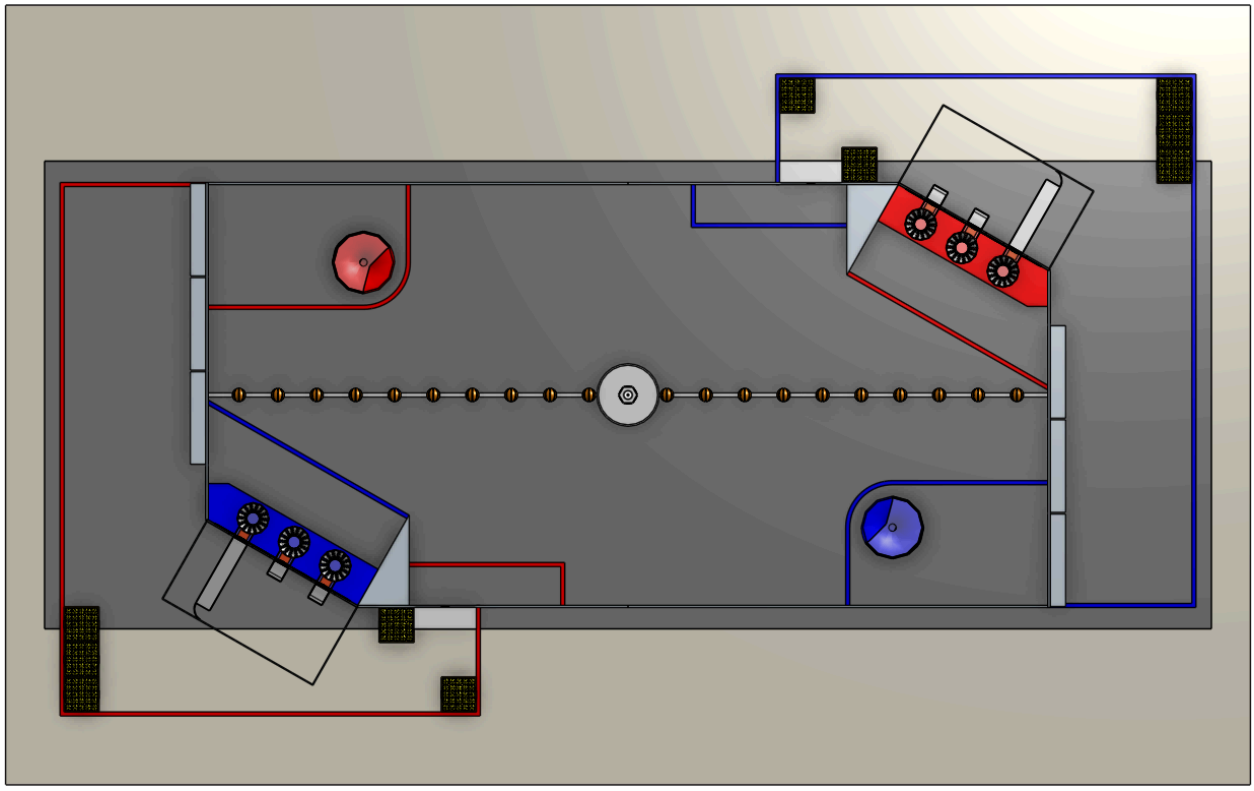
Task	Point Value	Point Value (Bonus Mode)
Dunk a BASKETBALL in the LOW HOOP	2 points each	4 points each
Dunk a BASKETBALL in the MIDDLE HOOP	5 points each	10 points each
Dunk a BASKETBALL in the HIGH HOOP	8 points each	16 points each
Score an ELECTROLYTE in their ALLIANCE DUNK TANK during the ENDGAME	⅓ point each	--
Completely FILL a HOOP STACK by the end of the match	20 points per HOOP STACK	--

The Arena

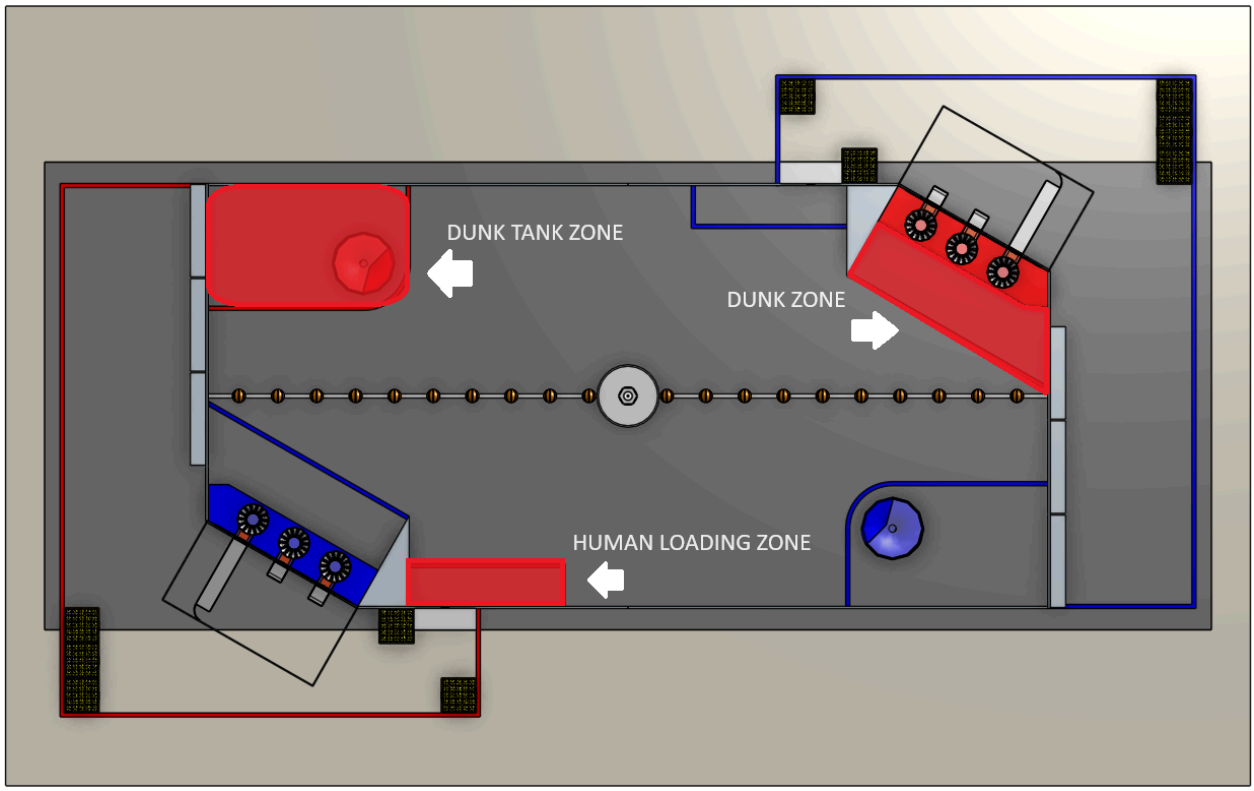


The field for *WCP DunkTank* is a 27’ wide by 54’ long carpeted area bounded by GUARDRAILS, ALLIANCE WALLS, and HOOP WALLS. The arena contains gameplay elements for *WCP DunkTank* such as the field, the HUMAN PLAYER STATIONS, the CORRALS, and the ELECTROLYTE BUCKETS.

Zones



Top Down view of the Arena



All zones are areas with an infinitely tall height to define the volume.

FIELD: The area defined by the GUARDRAILS, ALLIANCE WALLS, and HOOP WALLS.

DUNK TANK ZONE: The area defined by a GUARDRAIL, the ALLIANCE WALL, and alliance-colored tape that contains the alliance-colored DUNK TANK.

DUNK ZONE: The area defined by the HOOP BATTER, the ALLIANCE WALL, alliance-colored tape, and the ZONE DIVIDER.

HUMAN LOADING ZONE: The area defined by the ZONE DIVIDER, alliance-colored tape, and a GUARDRAIL that lies adjacent to the INBOUNDING BOX.

Game Elements

BASKETBALLS



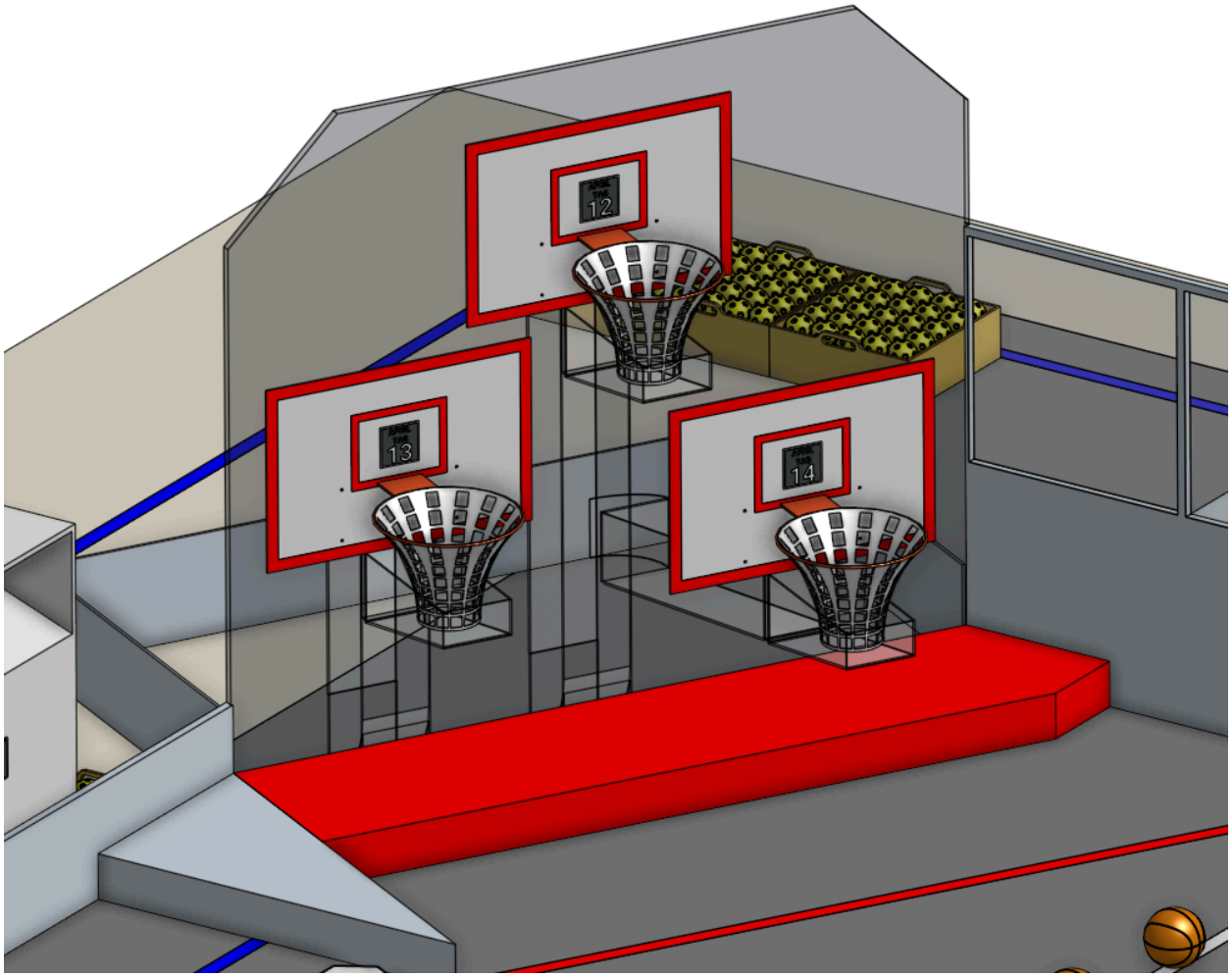
WCP DunkTank is played with standard Size 7 Basketballs, inflated to a circumference of 29.5”, or a diameter of roughly 9.39”. A match of *WCP DunkTank* utilizes 30 BASKETBALLS, with 20 starting the match evenly spaced along the white centerline and 5 positioned at each INBOUNDING BOX. Before the start of the match, each team may choose to pre-load their ROBOT with 1 BASKETBALL, taken from the 5 stored in their ALLIANCE INBOUNDING BOX.

ELECTROLYTES



ELECTROLYTES are 5” diameter wiffle balls reminiscent of the “fuel” game piece from the 2017 FRC game, FIRST Steamworks. In *WCP DunkTank*, ELECTROLYTES are used to activate BONUSES during the TELEOPERATED PERIOD and to score points during the AUTONOMOUS MODE and the ENDGAME. A match of *WCP DunkTank* contains 500 ELECTROLYTE game pieces, which start the match in containers outside of the field called ELECTROLYTE BUCKETS. Each ELECTROLYTE BUCKET holds 50 ELECTROLYTES, and each alliance begins the match with 5 ELECTROLYTE BUCKETS. During the course of the match, HUMAN PLAYERS may enter ELECTROLYTES into the field by pouring out the contents of ELECTROLYTE BUCKETS into the INBOUNDING BOXES. Once entered into the field, ELECTROLYTES are expected to stay inside the field for the duration of the match. Before the start of the match, each team may choose to pre-load their ROBOT with 1 ELECTROLYTE BUCKET’s worth of ELECTROLYTES (50) if they do not pre-load a BASKETBALL.

HOOPS



HOOPS are some of the primary scoring elements in *WCP DunkTank*. There are three HOOPS on each HOOP WALL: A LOW HOOP, a MIDDLE HOOP, and a HIGH HOOP, with ground to RIM heights of 36”, 54”, and 90”, respectively.

During the TELEOPERATED PERIOD, when a BASKETBALL enters a HOOP, it rolls through the HOOP WALL, passes the score count sensor, and falls into a HOOP STACK, which stores the BASKETBALL until an ALLIANCE STACK CLEAR is enabled. This means when a BASKETBALL is scored in TELEOPERATED, it is stored in the HOOP STACK and cannot be re-entered into play until the ALLIANCE STACK CLEAR conditions are met (explained in the DUNK TANKS section). Each HOOP STACK can store a different number of counted game pieces, regardless of its

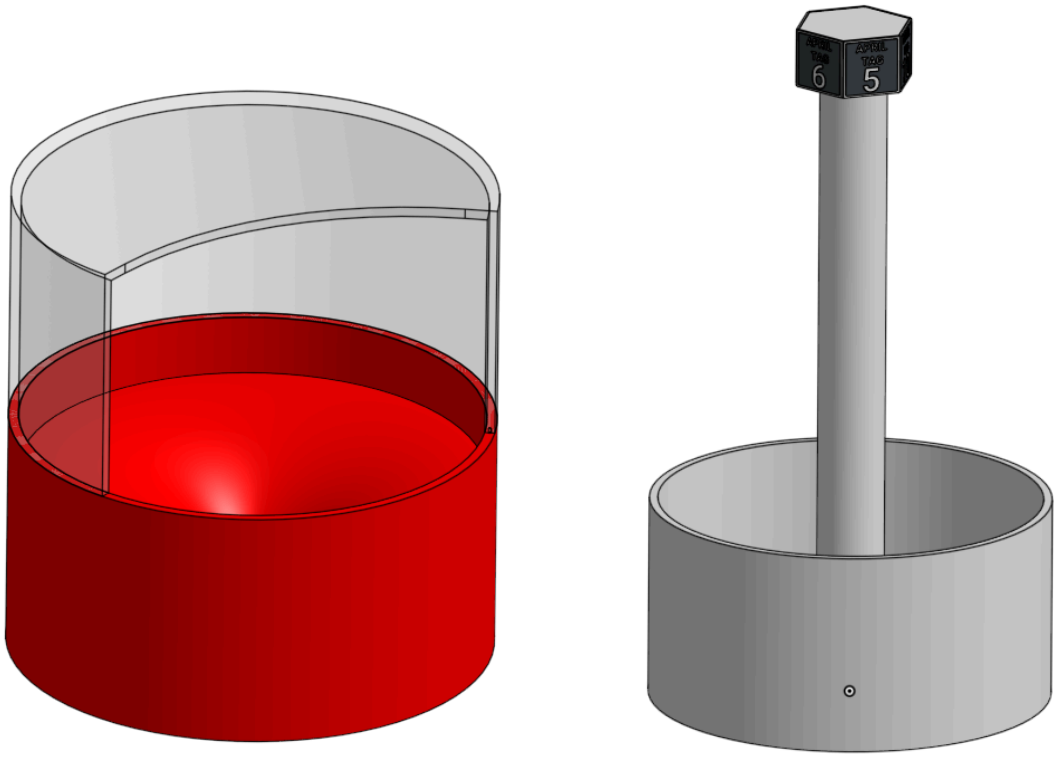
physical size. The LOW HOOP STACK can hold 6 BASKETBALLS, the MIDDLE HOOP STACK can hold 4 BASKETBALLS, and the HIGH HOOP STACK can hold 2 BASKETBALLS. Any additional BASKETBALLS scored in a HOOP that has already filled its HOOP STACK capacity will not count for any points. Once a HOOP STACK is full, an ALLIANCE STACK CLEAR must be enabled in order to continue scoring BASKETBALLS in that HOOP STACK. Furthermore, during the TELEOPERATED PERIOD the HIGH HOOP STACK will not begin counting, storing, or scoring BASKETBALLS until the LOW HOOP STACK and MIDDLE HOOP STACK are full.

During the AUTONOMOUS PERIOD, all HOOPS can be scored in regardless of STACK fill level.

When an ALLIANCE STACK CLEAR is enabled, all BASKETBALLS stored in HOOP STACKS will automatically be ejected from the HOOP STACKS into the CORRAL. While the BASKETBALLS will be removed from the HOOP STACKS, the points earned for the scored BASKETBALLS will remain.

At the end of the match, an ALLIANCE earns points for each HOOP STACK with a full capacity.

DUNK TANKS



DUNK TANKS are cylindrical containers designed to hold ELECTROLYTES. There are two kinds of DUNK TANKS: ALLIANCE DUNK TANKS and a BONUS MODE DUNK TANK, each with their own functions and effects. All DUNK TANKS are nominally 4’ in diameter with an opening 2’ off the ground at their top.

There are 2 ALLIANCE DUNK TANKS on the field, one red and one blue. These DUNK TANKS are located on opposite ends of the field in their respective ALLIANCE’S DUNK TANK ZONE. These DUNK TANKS have a large plastic lid on top with an opening that spans 120 degrees towards the center of the field. The inside of ALLIANCE DUNK TANKS is a funnel that exits to a chute behind the DUNK TANK where ELECTROLYTES can be re-entered back onto the field once the STACK CLEAR condition is met. An ALLIANCE DUNK TANK can comfortably hold 150 ELECTROLYTES in its base tank, with the ability for more to be stacked on top.

During the AUTONOMOUS PERIOD, ELECTROLYTES scored in an ALLIANCE DUNK TANK will earn points for each ELECTROLYTE scored. ELECTROLYTES scored in an ALLIANCE DUNK TANK during AUTO do not remain in the DUNK TANK.

In the TELEOPERATED PERIOD, ELECTROLYTES are not worth any points, but instead contribute towards the ALLIANCE STACK CLEAR effect. During TELEOPERATED, the exit chute is blocked off, preventing ELECTROLYTES from leaving the tank. Once 100 ELECTROLYTES are scored in an ALLIANCE DUNK TANK (measured by weight), the ALLIANCE STACK CLEAR effect is automatically activated, which will release all game pieces from HOOP STACKS and from the affected ALLIANCE DUNK TANK. The red ALLIANCE DUNK TANK affects the red ALLIANCE game elements and the blue ALLIANCE DUNK TANK affects the blue alliance game elements.

During the ENDGAME, the ALLIANCE STACK CLEAR effect is disabled and the exit chute in the ALLIANCE DUNK TANKS are opened for the remainder of the match. Additional ELECTROLYTES scored into the ALLIANCE DUNK TANKS during the ENDGAME will be worth points.

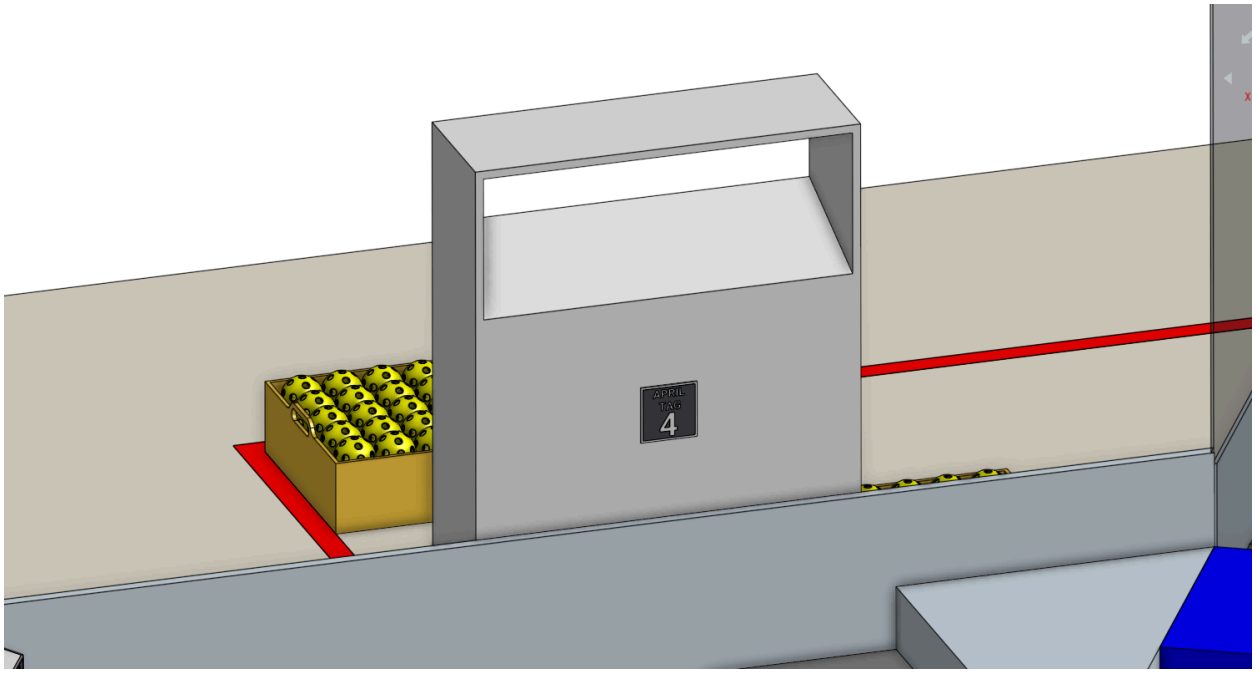
The BONUS MODE DUNK TANK is a white DUNK TANK located at the center of the field. At the center of the BONUS MODE DUNK TANK is a foam-covered pole 8” in diameter that extends 83” above the ground and contains 6 unique APRIL TAGS in a hexagon formation. The inside of the BONUS MODE DUNK TANK is a singular enclosed pocket where all ELECTROLYTES scored into the BONUS MODE DUNK TANK will stay for the duration of the match. The BONUS MODE DUNK TANK can comfortably hold 300 ELECTROLYTES. Every 100 ELECTROLYTES scored in the BONUS MODE DUNK TANK will give each ALLIANCE a BONUS MODE, which they can activate at a time of their choosing using a big red button behind their ALLIANCE WALL. Up to 3 BONUS MODES can be earned per ALLIANCE throughout the course of the match. BONUS MODE lasts for 10 seconds.

ELECTROLYTE BUCKETS



ELECTROLYTE BUCKETS are containers located outside of the field that HUMAN PLAYERS can use to enter ELECTROLYTES into the field through the INBOUNDING BOX. Each ELECTROLYTE BUCKET holds 50 ELECTROLYTES stacked in a 2 tall, 5x5 grid. 5 ELECTROLYTE BUCKETS will start the match behind each ALLIANCE WALL. HUMAN PLAYERS must introduce all ELECTROLYTES onto the field before the final 60 seconds of the match. The ELECTROLYTE BUCKETS themselves should not be entered into the field. They are not a game piece.

INBOUNDING BOXES



The INBOUNDING BOXES are game elements located outside of the field barrier that HUMAN PLAYERS use to introduce BASKETBALLS and ELECTROLYTES to the field. There is one INBOUNDING BOX along the side of each ALLIANCE WALL which the ALLIANCE's HUMAN PLAYERS can use at any point during the match. INBOUNDING BOXES are located in the HUMAN LOADING ZONE. The lower lip of the INBOUNDING BOX sits 48" above the floor of the field.

The Game

Fouls

- Each FOUL adds 10 points to the opposing ALLIANCE'S score
- Each TECHNICAL FOUL adds 20 points to the opposing ALLIANCE'S score
- Strategies or intended actions that result in a DQ will DISQUALIFY a team's ROBOT from the CADATHON

Game Rules

G01: When placed on the field, each ROBOT must be:

A) Confined to its STARTING CONFIGURATION
B) Positioned fully within their ALLIANCE'S DUNK ZONE or DUNK TANK ZONE
C) Fully supported by the floor of the field

G02: Strategies aimed at forcing a FOUL or a TECHNICAL FOUL on the opposing ALLIANCE are not allowed.

Violation: TECHNICAL FOUL, forced FOUL not called

G03: Scores are final when everything on the FIELD comes to a stop, or 5 seconds after the match ends, whichever comes first.

G04: ROBOTS may not contact an OPPOSING ROBOT on the opposite side of the white CENTER LINE during the AUTONOMOUS PERIOD.

Violation: TECHNICAL FOUL for the off-sides ROBOT, DQ if strategic

G05: ROBOTS may not use an extension outside of their FRAME PERIMETER to contact an opposing ROBOT inside their FRAME PERIMETER as part of normal conditions of play.

AKA: Do not strategically intend to extend into another ROBOT'S volume.

Violation: TECHNICAL FOUL for the initiator

G06: ROBOTS may not intentionally detach or leave parts on the FIELD.

Violation: TECHNICAL FOUL, DQ if match-affecting

G07: ROBOTS may not extend out of the FIELD. Brief, incidental incursions will not be penalized.

Violation: FOUL, TECHNICAL FOUL if contact is made outside of the FIELD

G08: A ROBOT'S horizontal dimensions may never exceed 18" beyond its FRAME PERIMETER.

Violation: FOUL per violation

G09: ROBOTS may not intentionally fall over.

Violation: FOUL

G10: Do not actively attempt to harm other ROBOTS.

Violation: DQ

G11: A pin cannot last longer than 5 seconds.

Info: A pin requires a ROBOT on one ALLIANCE to sandwich a ROBOT of another SQUAD between itself and another object in the FIELD.

Violation: TECHNICAL FOUL

G12: An ALLIANCE may not stop the flow of a match. Examples of this would be blockading narrow traveling areas, *blocking access to the opening of the DUNK TANKS, or withholding BASKETBALLS behind the ALLIANCE WALL and not returning them to the field upon a STACK CLEAR.*

Violation: TECHNICAL FOUL, DQ if strategic

G13: HUMANS may not extend into the FIELD. Accidental incursions will be met with a warning.

Violation: FOUL, TECHNICAL FOUL if repeated

G14: Regardless of who initiates contact, ROBOTS may not contact an opposing ROBOT when the opposing ROBOT is at least partially within their ALLIANCE DUNK ZONE, *fully* ALLIANCE DUNK TANK ZONE, or ALLIANCE HUMAN LOADING ZONE.

Violation: FOUL per contact.

G15: A ROBOT may only be in possession of 1 BASKETBALL at a given time. There is no limit on the number of ELECTROLYTES that can be in a ROBOT'S possession.

Violation: TECHNICAL FOUL

G16: BASKETBALLS and ELECTROLYTES may only be entered into the FIELD via the slot in the INBOUNDING BOXES.

Violation: TECHNICAL FOUL

G17: ROBOTS may not use BASKETBALLS or ELECTROLYTES to ease or amplify any of the challenges in *WCP DunkTank*.

Violation: DQ

G18: A ROBOT must be within its' ALLIANCE DUNK ZONE to score a BASKETBALL.

Violation: All ALLIANCE HOOP STACKS are cleared, TECHNICAL FOUL

G19: BASKETBALLS released from ROBOT possession below the height of the HOOP they enter are ineligible for points.

Aka: You have to dunk the BASKETBALLS in, no launching

Violation: 2 TECHNICAL FOULS, and the HOOP STACK will be ineligible for BONUS MODE and the BONUS at the end of the match.

G20: ROBOTS may only extend above the maximum starting height of 42 inches while within their ALLIANCE DUNK ZONE.

Violation: Foul per instance

G21: Upon the activation of a STACK CLEAR, alliance HUMAN PLAYERS must return all BASKETBALLS into the field within 30 seconds.

Violation: FOUL per BASKETBALL remaining, TECHNICAL FOUL per BASKETBALL remaining if strategic

ROBOT Rules

R01: The ROBOT must fit the following size constraints:

A) The external FRAME PERIMETER cannot be greater than 120 inches
B) The ROBOT may not extend more than 18 inches outside of its FRAME PERIMETER in any horizontal direction

R02: In its STARTING CONFIGURATION, the ROBOT must fit entirely within its frame perimeter and be no taller than 42 inches.

R03: The ROBOT weight (excluding battery and bumpers) may not exceed 125 lbs.

R04: No cleats / stabbing into the carpet to gain traction.

R05: ROBOTS may not be made of any parts that disrupt or interfere with other ROBOTS.

R06: FRC 2024 motor, electrical, and pneumatic rules apply.

R07: FRC 2023 bumper rules apply, alternatively teams can use the new 2025 bumper rules (please state if you are doing this in your documentation)

The Tournament

T01: Ranking Points are calculated as follows:

A) Each win is +2 RP
B) Each tie is +1 RP
C) Each loss is +0 RP
D) Ending a match with all 3 HOOP STACKS filled is +1 RP
E) Activating 3 ALLIANCE STACK CLEARS OR 1 ALLIANCE STACK CLEAR and 1 BONUS MODE is +1 RP

T02: Seeding will be determined in the following order:

A) Highest Ranking Points
B) Most ALLIANCE STACK CLEARS
C) Most HOOP STACKS filled at the end of matches
D) Most BONUS MODES activated
E) The team with the most corgis